

# A mirror step becomes a phase; fields add as arrows, so light can cancel

1 the surface moves by  $h$

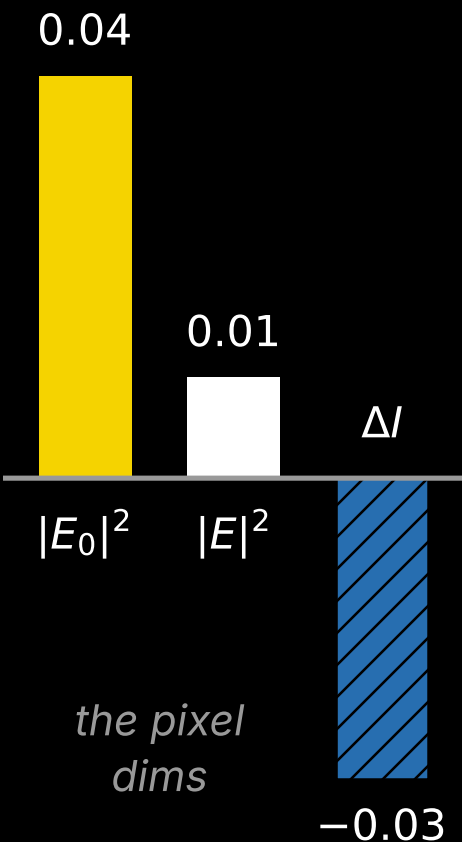
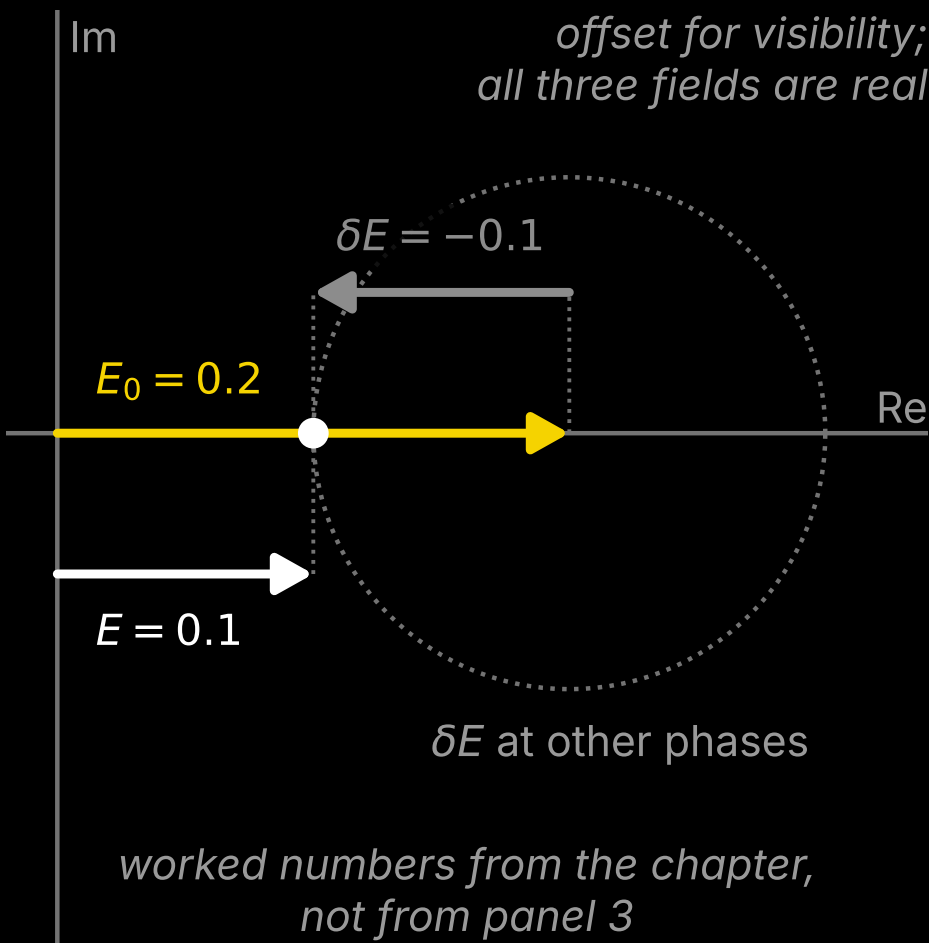
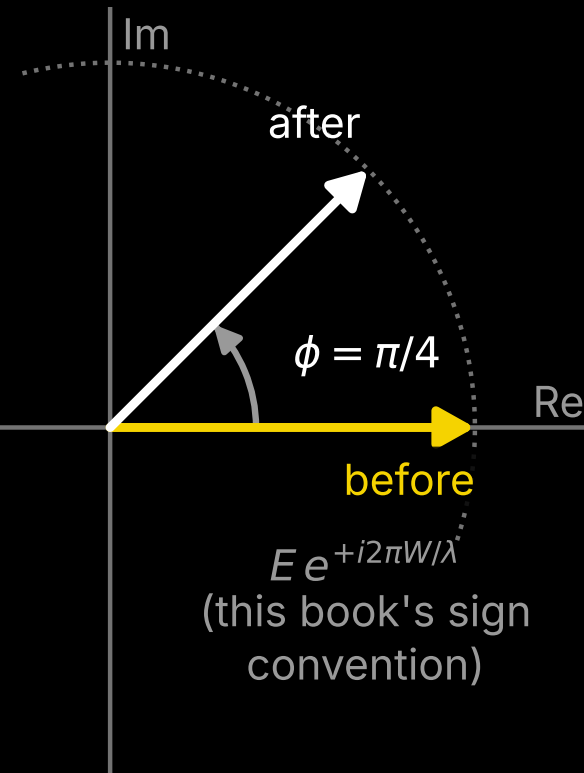
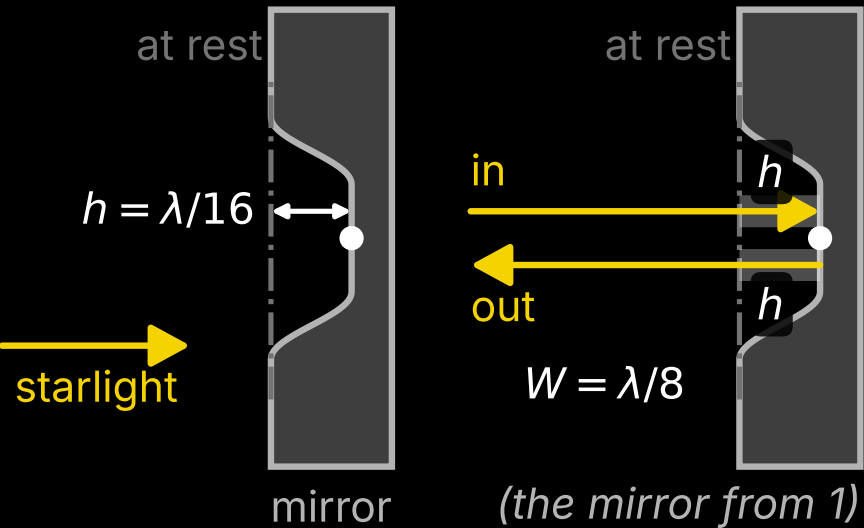
2 the path grows by  $W = 2h$

3 the phase turns by  $2\pi W/\lambda$

4 at one pixel, the fields add as arrows

*pupil point  $\rightarrow$  image pixel: the pixel's field is a sum of many turned arrows*

5 brightness = (arrow length)<sup>2</sup>



$$\Delta I = 2 \operatorname{Re}(E_0^* \delta E) + |\delta E|^2 = -0.04 + 0.01 = -0.03$$

*schematic, not to scale*